Dominick Cole
HUMN-391-02
16 April 2026

Project #1: Personal Statement

I was ten years old the first time I flew on a plane. Somewhere along the route to Florida, I had my face pressed against the window, watching the wings flex up and down with the turbulence. I did not understand how something that massive could bend without breaking, or how something as heavy as an airplane could stay afloat for what seemed like millions of miles. After we landed, the pilot let me sit in the cockpit, and I remember seeing all the switches and gauges, wondering how someone could design it all. I did not have the vocabulary at the time, but what I felt in that moment was awe at all the complex thought, execution, physics, and design that go into creating an airplane. That experience is the reason I chose engineering. It was not about job security or a high salary. It was a genuine want to understand how complex systems work and eventually become one of the people who build them. My goal is to work as an engineer in the aerospace industry, contributing to the design and development of commercial aircraft. Through both my academic coursework and co-op experience in the automotive industry, I have spent the last several years turning that enthusiasm into competence, and I believe that combination is what positions me to contribute meaningfully to this field.

That curiosity carried me through high school and into Kettering University, where I enrolled as a mechanical engineering student. Through Kettering's co-op program, I began working at Martinrea International, where my group focused on steel structural and chassis components, such as control arms and subframes. Using SolidWorks, CATIA, and NX, I have gained direct experience in design and product engineering as well as testing and defect investigation. One of the more hands-on projects I worked on involved investigating necking defects on hydroformed rails. I measured the wall thickness in and out of the affected zones using a point micrometer and performed cut-and-etch analysis to help determine where the material was failing. This kind of material failure analysis and dimensional inspection translates directly to the aerospace industry, where engineers perform similar evaluations on aircraft structural components to ensure they meet safety and performance standards. This work required attention to detail and gave me an appreciation for how closely manufacturing processes and material behavior are connected.

On the design side, I took on a project that required me to think beyond modeling a part. Our team was experiencing issues with a bracket weld on a subframe, and a weld engineer needed a way to visually locate the trim line before the bracket was welded on to investigate the problem. I designed a locating jig optimized for 3D printing that indexed off three points on the subframe and included a line that indicated the trim line location. It was a fairly complex geometry problem, but it required me to understand the problem clearly, think about how the tool would actually be used on the floor, and ensure my design could be 3D printed. I also contributed to benchmarking efforts, tearing down and comparing competitor parts to evaluate design choices and identify opportunities for our parts to improve. Alongside these co-op experiences, coursework in materials science and dynamic systems reinforced the analytical thinking I was developing on the job, giving me both a theoretical and practical foundation for solving structural problems. These experiences taught me that I do not shy away from problems I don't immediately understand and that I'm willing to ask questions when I need to. These are qualities I developed not in a classroom alone, but through the combination of academic study and real work on a lab floor.

At the same time, as I became more involved in the automotive industry, I began to feel a disconnect. The engineering of it all was what I expected, as I was constantly learning and expanding my skill set, but I had started to lose sight of the specific ambition that had gotten me into the field in the first place.

The moment that brought everything back into focus happened during my junior year in a fluid mechanics lab. We were running a scaled-down airfoil through a wind tunnel, measuring lift and drag coefficients at different angles of attack. On paper, it was a typical lab assignment. But standing there watching the airflow interact with the wing and seeing the lift data update in real time, I felt something I hadn't in a long time. The lift on that airfoil was governed by the same physics that had kept the plane airborne all the way back when I was ten. The difference was that now I had spent three years learning the math and physics behind it. For the first time in a while, the work felt personal again.

That lab did not change my direction so much as it reminded me that I already had one. It clarified something I had been struggling to wrap my head around. It clarified that mechanical engineering was never a detour from aerospace; it was the foundation for it. Every course I had taken in statics, dynamic systems, thermodynamics, and materials science was building toward the kind of work I originally set out to do. The wind tunnel lab was the moment where all of those pieces came together, and I could see the full picture. It revived my commitment to pursuing aerospace engineering as a career and gave me a sense of urgency about making that transition.

Looking back, I think the most valuable thing about my path so far is that it has not been a straight line. I went into college already knowing what I wanted, lost sight of it for a while during my co-op rotations and coursework, and then found my way back through a single lab experience that reconnected me with my original motivation. That process was frustrating at the time, but it taught me something I do not think I could have learned any other way. Without the technical skills and self-awareness to back it up, passion is just enthusiasm. I have spent the last several years turning that enthusiasm into competence, and I believe that combination is what will make me effective in the aerospace industry.

As I approach graduation, my goal is to secure an entry-level position in the aerospace industry, particularly in the analysis and design of commercial aircraft structures. The industry is in the middle of a shift toward lightweight composite materials, more fuel-efficient airframes, and sustainable aviation technologies, and I want to be part of the teams driving that work. The CAD proficiency, structural investigation skills, and hands-on problem-solving I developed at Martinrea are directly applicable to the work aerospace engineers do every day, from evaluating material performance to designing components under tight constraints. My academic journey at Kettering has given me the educational foundation I need to succeed. And my personal journey has given me something that I think matters just as much. It has given me a clear sense of why I am pursuing this work and the persistence to see it through. I am not simply looking for a job in the aerospace industry. I am looking to build a career in a field that has motivated me since I was a kid, and I am ready to bring everything I have learned to that effort.